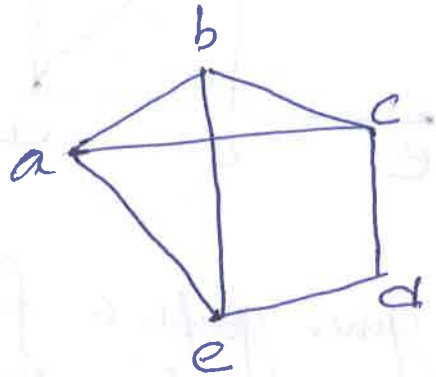
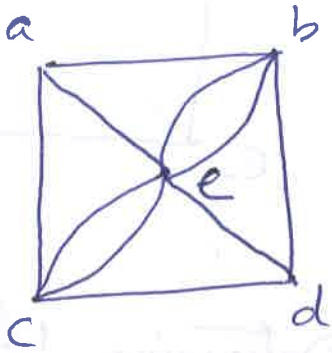
Tutorial - 6

Q Determine whether the given graph has an Euler Circuit. Construct such a Circuit. If no Euler Circuit exists, determine whether the graph has an Euler path.

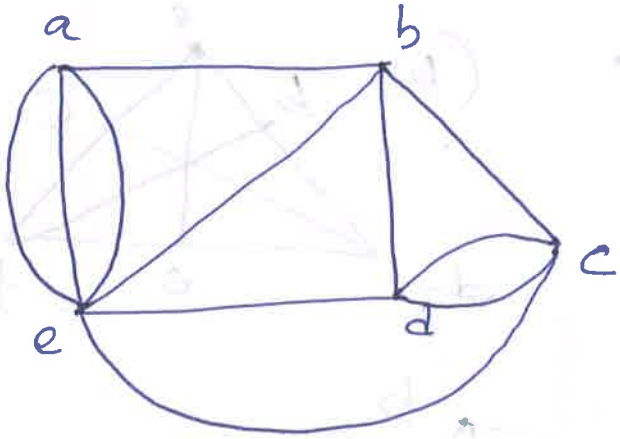
(a)



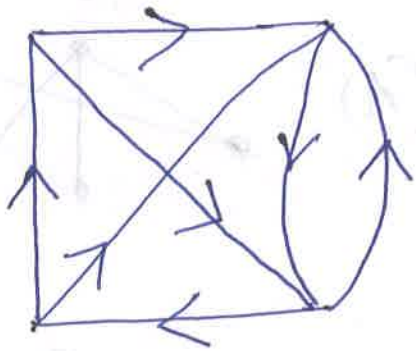
(b)



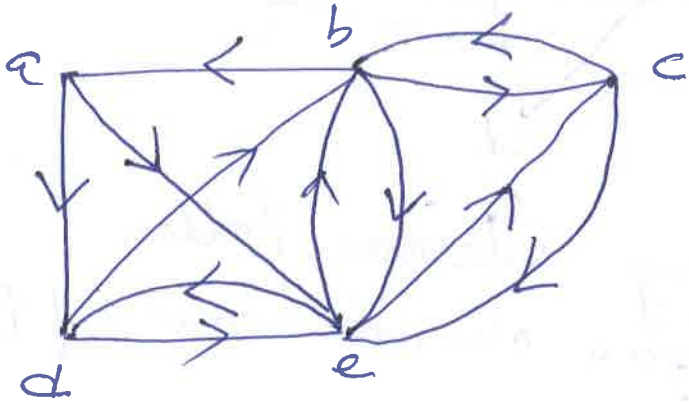
(c)



(D)

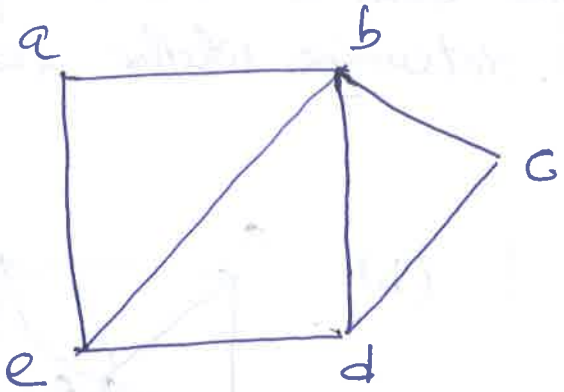


(e)

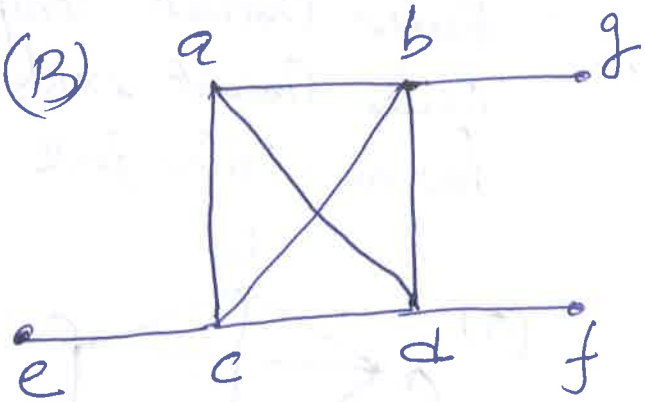


Q2 Determine whether the given graph has a hamiltonian Circuit. If it does find such a Circuit.

(A)

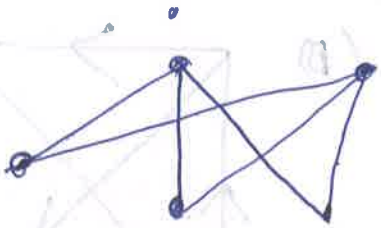


(B)

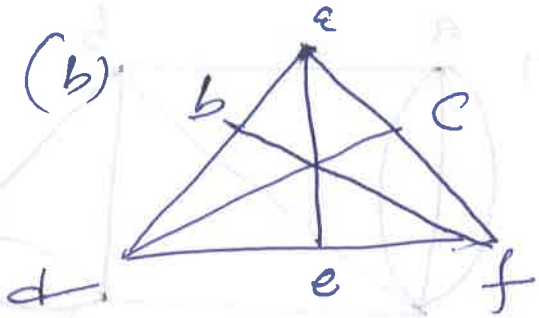


Q3 Determine whether the given graph is planar. If so, draw it so that No edges cross.

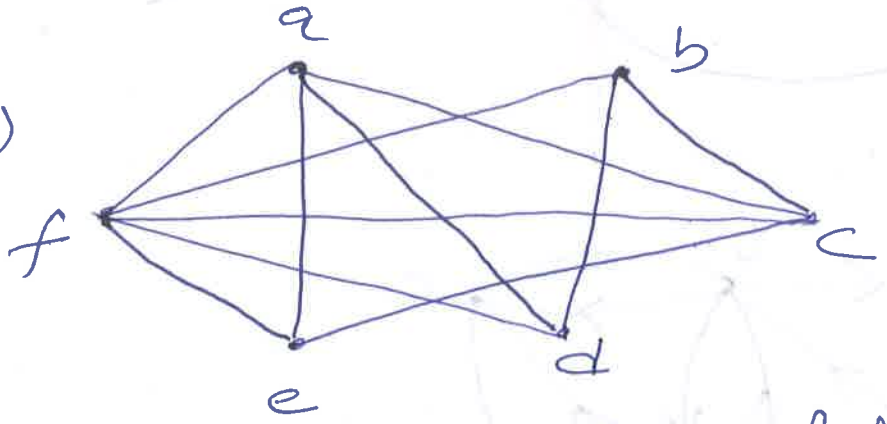
(A)



(b)



(e)



Q4. Explain Travelling Salesman Problem.

Q5. Which graph has a chromatic number of 1.

Q6 Suppose that Someone starts a chain letter. Each person who receives the letter is asked to Send it on to four other people. Some people do this but others do not Send any letters. How many people have Seen the letter including the first person if No one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not Send it out. How many people Sent out the letters?

Q7. How many edges does a tree with 10,000 vertices have?

Q8. How many different ISOMERS do these Saturated hydrocarbons have?

(a) C_4H_{10} (Butane)

(b) C_3H_8

Q9. A Labeled tree is a tree where each vertex is assigned a Label. Two labeled trees are Considered isomorphic when there is an isomorphism between them that preserves the label of vertices. How many non isomorphic trees are there with three vertices labeled with different integers from the set $\{1,2,3\}$?

Q10 Consider a Complete undirected graph with vertex Set $V = \{0,1,2,3,4\}$. Entry W_{ij} in the matrix W below is the weight of the edge $\{i,j\}$.

$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix}$$

(a) What is the minimum possible weight of a Spanning tree T in this graph Such that vertex 0 is a leaf Node in the tree T?

(b) What is the minimum possible weight of a path P from vertex 1 to vertex 2 in this graph Such that P Contains at most 3 edges?